

Project Design Phase-II

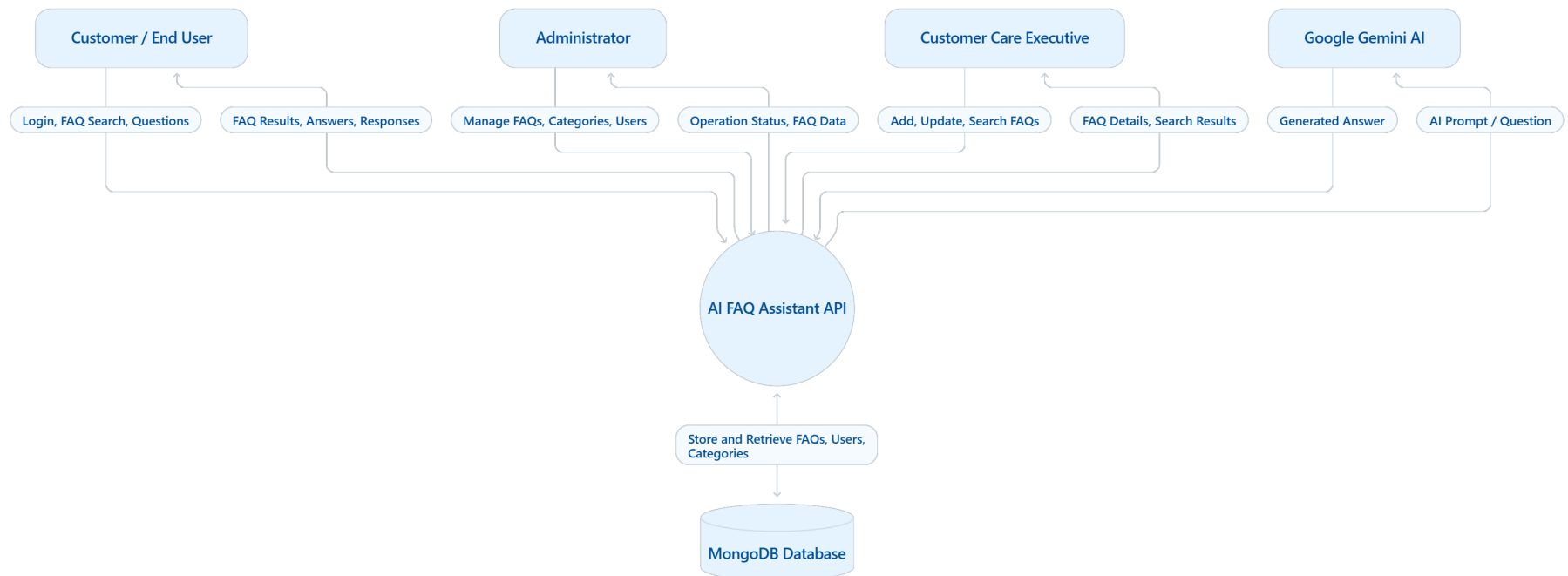
Data Flow Diagram & User Stories

Project Name	AI FAQ Assistant API
Date	30TH SEPTEMBER 2026
Team ID	SWTID-2026-3342

2. Data Flow Diagrams (DFD)

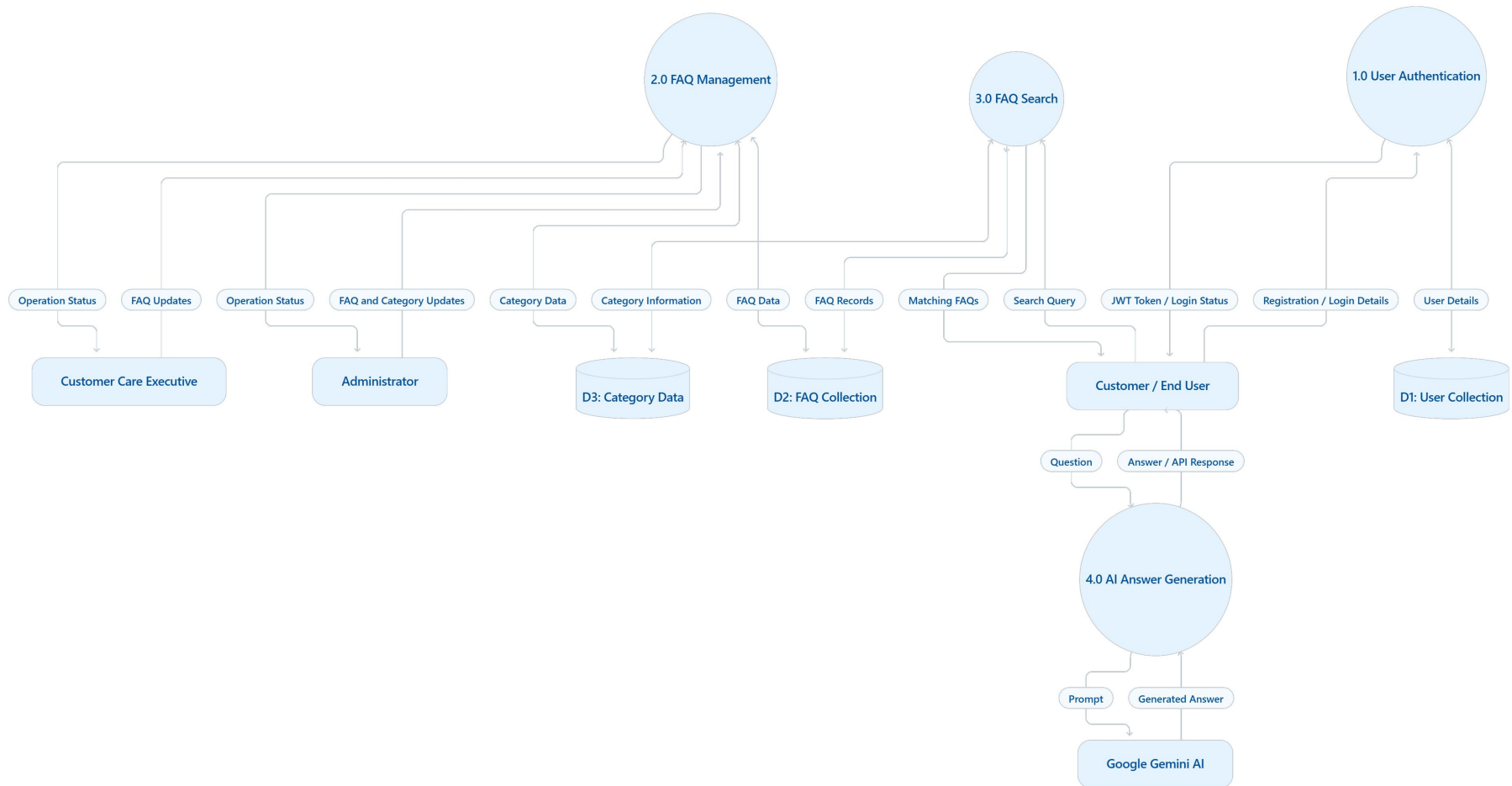
2.1 DFD Level 0 – Context Diagram

Description: The Level 0 Data Flow Diagram represents the AI FAQ Assistant API as a single system and shows how external users and services interact with it. Users can register, log in, search FAQs, and receive answers. Administrators and customer care executives can manage FAQs, while the system stores data in MongoDB and uses Google Gemini AI to generate AI-assisted responses.



2.2 DFD Level 1 – Detailed Data Flow

Description: The Level 1 DFD breaks the AI FAQ Assistant API into its major processes: user authentication, FAQ management, FAQ search, and AI-powered answer generation. The Express.js API receives requests, validates them, interacts with MongoDB through Mongoose, and returns responses in JSON format. JWT is used for authentication, and bcrypt is used for password hashing.



2.3 Main Data Stores

Data Store	Description
D1 – User Collection	Stores registered user details and securely hashed passwords.
D2 – FAQ Collection	Stores FAQ questions, answers, and related information.
D3 – Category Data	Stores FAQ categories used to organize and filter FAQs.

Note: The diagrams show the main logical data flows. The user, FAQ, and category data are stored in MongoDB. Authentication uses JWT and bcrypt, and AI-generated answers are obtained through the Google Gemini API.

3. User Stories

The following user stories use the template columns: User Type, Functional Requirement (Epic), User Story Number, User Story / Task, Acceptance Criteria, Priority, and Release.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web/Mobile User)	User Registration	USN-1	As a user, I can register for the application by providing my required details.	The user account is created successfully when valid details are provided. Invalid or incomplete details are rejected.	High	Sprint-1
Customer (Web/Mobile User)	User Login	USN-2	As a user, I can log in using my registered credentials.	Valid credentials allow access. Invalid credentials return an appropriate error message.	High	Sprint-1
Customer (Web/Mobile User)	Authentication	USN-3	As a user, I can access protected API features using an authentication token.	A valid JWT grants access to protected endpoints. Missing or invalid tokens are rejected.	High	Sprint-1
Customer (Web/Mobile User)	FAQ Search	USN-4	As a user, I can search for FAQs using keywords related to my question.	The API returns matching FAQs. If no results are found, an appropriate response is displayed.	High	Sprint-1
Customer (Web/Mobile User)	FAQ Browsing	USN-5	As a user, I can view available FAQs and their answers.	The API returns the requested FAQ details in JSON format.	High	Sprint-1
Customer (Web/Mobile User)	Category-Based Search	USN-6	As a user, I can find FAQs by selecting or specifying a category.	FAQs belonging to the selected category are returned. Invalid categories are handled appropriately.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web/Mobile User)	AI-Powered Answers	USN-7	As a user, I can ask a question and receive an AI-generated answer.	The question is sent to Google Gemini AI, and the generated answer is returned through the API.	High	Sprint-2
Customer (Web/Mobile User)	Error Handling	USN-8	As a user, I receive a clear message when my request cannot be processed.	Invalid input, unavailable resources, and service errors return appropriate error responses.	Medium	Sprint-2
Customer Care Executive	FAQ Management	USN-9	As a customer care executive, I can add new FAQs to help customers find answers.	Valid FAQ details are saved in MongoDB and can be retrieved through the API.	High	Sprint-2
Customer Care Executive	FAQ Management	USN-10	As a customer care executive, I can update existing FAQ questions and answers.	The selected FAQ is updated successfully, and the changes are reflected in subsequent retrievals.	High	Sprint-2
Customer Care Executive	FAQ Management	USN-11	As a customer care executive, I can delete outdated or incorrect FAQs.	The selected FAQ is removed, and the API confirms the operation.	Medium	Sprint-2
Administrator	Category Management	USN-12	As an administrator, I can create and manage FAQ categories.	Categories can be created, retrieved, updated, and managed through authorized API endpoints.	Medium	Sprint-2
Administrator	Access Control	USN-13	As an administrator, I can restrict FAQ management operations to authorized users.	Protected operations require valid authentication and appropriate access permissions.	High	Sprint-2
Administrator	FAQ Oversight	USN-14	As an administrator, I can view and maintain the FAQs stored in the system.	The administrator can retrieve FAQ records and perform permitted management operations.	Medium	Sprint-2
Administrator	API Reliability	USN-15	As an administrator, I can receive consistent API responses for successful and failed requests.	API responses use a consistent JSON structure and appropriate HTTP status codes.	Medium	Sprint-2

4. Summary

The AI FAQ Assistant API provides a centralized platform for managing frequently asked questions and delivering automated customer support. The system allows customers to search FAQs, browse categories, and obtain AI-generated answers. Customer care executives can maintain FAQ content, while administrators oversee FAQ and category management.

The system uses Node.js and Express.js to build RESTful APIs, MongoDB and Mongoose for data storage, JWT and bcrypt for authentication and password security, and Google Gemini AI for generating AI-assisted responses. The DFDs illustrate the movement of data between users, API processes, the database, and the AI service, while the user stories define the functional requirements and acceptance criteria for development.